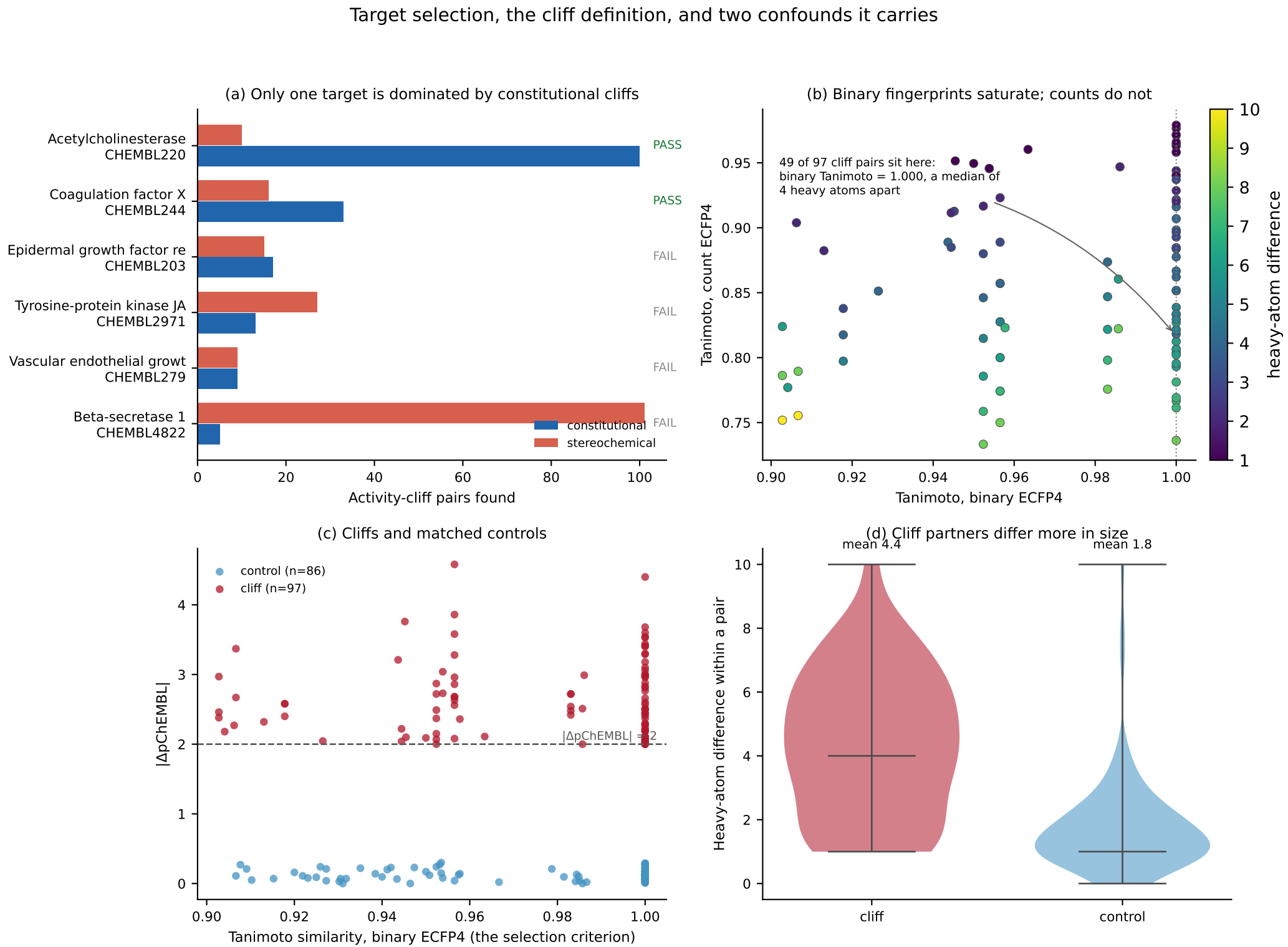
Target selection, the cliff definition, and two confounds it carries
(a) Only one target is dominated by constitutional cliffs
Acetylcholinesterase
CHEMBL220
Coagulation factor X
CHEMBL244
Epidermal growth factor re
CHEMBL203
Tyrosine-protein kinase JA
CHEMBL2971
Vascular endothelial growt
CHEMBL279
Beta-secretase 1
CHEMBL4822
PASS
PASS
FAIL
FAIL
FAIL
FAIL
constitutional
stereochemical
0
20
40
60
80
100
Activity-cliff pairs found
(b) Binary fingerprints saturate; counts do not
49 of 97 cliff pairs sit here:
binary Tanimoto = 1.000, a median of
4 heavy atoms apart
Tanimoto, count ECFP4
0.75
0.80
0.85
0.90
0.95
0.90
0.92
0.94
0.96
0.98
1.00
Tanimoto, binary ECFP4
heavy-atom difference
1
2
3
4
5
6
7
8
9
10
(c) Cliffs and matched controls
control (n=86)
cliff (n=97)
|ΔpChEMBL| = 2
|ΔpChEMBL|
0
1
2
3
4
0.90
0.92
0.94
0.96
0.98
1.00
Tanimoto similarity, binary ECFP4 (the selection criterion)
(d) Cliff partners differ more in size
mean 4.4
mean 1.8
Heavy-atom difference within a pair
0
2
4
6
8
10
cliff
control